

August 2, 2026

Dear Hiring Team,

I am writing to apply for the AI Engineer position at maru.ai. With a Master of Science in Computer Science from the University of Calgary and hands-on experience in machine learning, Python software development, model evaluation, and scalable data workflows, I am excited by the opportunity to help build AI capabilities for an early-stage company focused on practical product impact.

My graduate research and software work have centered on turning complex data problems into reliable technical systems. In my DEM super-resolution project, I trained and evaluated deep ResNet models in PyTorch to reconstruct 30m elevation data from 120m inputs, built preprocessing workflows for large geospatial datasets, and designed evaluation methods using statistical and terrain-specific metrics. This work required careful attention to model accuracy, generalization, data quality, computational cost, and reproducible experimentation.

I also developed a spatiotemporal Earth observation data system using C++ and Python, where I built encoding pipelines, optimized retrieval workflows, and improved processing performance with multithreading. The project strengthened my ability to design software that balances correctness, scalability, and performance. In addition, my backend development experience includes building Django REST APIs, working with PostgreSQL, implementing authentication and payment-service logic, and contributing to NLP and chatbot-related services.

The responsibilities of this role match the kind of work I enjoy: designing AI/ML models, evaluating new techniques, collaborating across product and engineering, and improving systems for accuracy, latency, and scalability. I bring strong Python experience, practical deep learning knowledge, C++ software engineering experience, and exposure to AWS-based data workflows through both research and teaching. I have not tried to present myself as a Go expert, but I am comfortable learning new engineering stacks quickly and applying them with care.

I also value the responsible-AI aspects of the role. My research has made me attentive to sampling choices, evaluation design, model transparency, and the ways technical decisions can affect downstream reliability. I would bring that same care to maru.ai's work around AI architecture, deployment practices, explainability, and ethical use.

Thank you for considering my application. I would welcome the opportunity to discuss how my machine learning, software engineering, and research-to-product experience can contribute to maru.ai's AI engineering work.

Sincerely,
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